

Crystal types

There are crystals all around us. Most rocks are made of them, and most sandy beaches are made of billions of broken quartz crystals. Even the sugar and salt in your kitchen are crystals. Some of the most spectacular are mineral crystals that grow in underground cavities, sometimes reaching enormous sizes.

Quartz

Abundant and beautiful

Main ingredients Silicon, oxygen

Crystal type Trigonal

Hardness on 1–10 scale 7

White quartz is pure silica—the compound of silicon and oxygen that we use to make glass. It is one of the most common minerals in continental rocks, where the crystals often form clusters inside rock cavities. Typical quartz crystals are six-sided columns topped with six-sided pyramids.

► AMETHYST

This violet form of quartz owes its beautiful color to impurities in the mineral, including traces of iron.

Beryl

Precious emerald

Main ingredients Beryllium, aluminum, silicon, oxygen

Crystal type Hexagonal

Hardness on 1–10 scale 7.5–8

This hard mineral forms crystals in rocks that have been altered deep beneath the ground. Some crystals can grow larger than cars, but the most valuable are smaller green emeralds, as below. Beryl also occurs in a blue-green form called aquamarine.

